

ONE-ON-ON EXCHANGES BETWEEN TEACHERS AND STUDENTS

The following are sample conversations between individual students and Teachers A, B, C, and D. Each relates to the question (below) from the Citizen Math lesson Big Foot Conspiracy in which students apply unit rates to explore pricing conventions.

Q2. For each size, calculate how much a pair of All Stars costs per ounce. Based on this, who would you say is paying the most for shoes and do you think this is fair? Explain.

SIZE	4	6	8	10	12	14	16
PRICE	\$50	\$50	\$50	\$50	\$50	\$50	\$50
WEIGHT (OZ)	19.6	22.6	24.2	26	30.2	31.4	37.6
COST PER OUNCE	---	---	---	---	---	---	---

Teacher A

T: What are we trying to find?

S: The price per ounce of each size.

T: What does "cost per ounce" really mean?

S: The price for each ounce of shoe.

T: For size 4, how do we calculate the cost per ounce?

S: ...I'm not sure.

T: For size 4, does each shoe cost \$1?

S: No.

T: How do you know?

S: If it did, 19.6 ounces would cost almost \$20, not \$50.

T: Does each ounce cost \$3?

S: No. If it did, the pair would cost \$60.

T: Okay. So how much are you paying per ounce? See if you can figure this out for size 4. I'll come back in a few minutes to check in.

Teacher B

T: What are we trying to find?

S: The price per ounce of each size.

T: And how can you do that?

S: I'm not sure.

T: What do you know about the shoes?

S: They're made by Converse. They come in different sizes. Each size costs the same but weighs a different amount.

T: Can you use this to figure out the cost per ounce?

Teacher C

T: What are we being asked to find?

S: The price per ounce of each size.

T: For size 4, what is the price of the pair, in dollars?

S: \$50.

T: For size 4, what is the weight of the pair, in ounces?

S: 19.6 ounces.

T: What number do you get when you divide 50 by 19.6?

S: 2.55.

T: And what are the units of that?

T: When you divide dollars by ounces, you get "dollars per ounce."

T: So what are the units of 2.55?

S: Dollars per ounce?

T: Right. So what is the unit rate for size 4?

S: 2.55 dollars per ounce?

T: Good. Now do this for size 16. What do you get?

S: \$1.33 per ounce.

Teacher D

T: What are we trying to find?

S: The price per ounce of each size.

T: How do we determine the cost per ounce?

S: I'm not sure.

T: Let's imagine a different scenario. Imagine a 6-pack of soda costs \$12. How much would you be paying for each can?

S: \$2.

T: How did you get that?

S: I divided \$12 by 6 cans.

T: Now imagine that 20 pounds of candy costs \$10. How much are you paying for each pound?

S: \$10 divided by 20 is $\frac{1}{2}$, so \$0.50.

T: Now let's return to the shoes. For a size 4, how much does the pair cost and how many ounces does it contain?

S: \$50 for 19.6 ounces.

T: So how much are you paying for each ounce?

S: $50 \div 19.6 \text{ ounces} = 2.55$, so \$2.55 per ounce.

T: How does this compare to size 16?

S: $50 \div 37.6 \text{ ounces} = 1.33$, so \$1.33 per ounce. The large shoe costs less per ounce than the small shoe.